

		$= 330 / 0.24$	M1
		$= 1400(1375 \text{ Hz})$	A1
	(iii)	$t = d / v$	
		$= 12 / 0.75$	M1
		$= 16 \text{ s}$	A1
	(iv)	The actual wavelength was longer due to end corrections.	B1
5	(a)	(i) When p.d. V increases, current I through the filament lamp also increases, with <u>decreasing I-V ratio.</u>	B1
		As <u>I increases</u> , power dissipated increases ($P = I^2 R$). Heat is generated and hence <u>equilibrium temperature increases</u> . (This is because as electrons drift through the metal, they collide with the metal lattice, transferring energy to them)	B1
		As equilibrium temperature increases, the <u>lattice ions vibrate more vigorously, hindering the flow of 'charge carriers'</u> , therefore increasing the resistance.	B1
	(ii)	$E = I (3R + 2r)$ $8.7 = 0.30 (3 (2.5/0.3) + 2r)$ $r = 2.0 \Omega$	M1 A1
	(iii)	Straight line passing through (2.0, 0.10) and (4.0, 0.20)	B1
(b)	(i)	$V_{AJ} = E_2$	C1
		$V_{AJ} = \frac{\left(\frac{63.1}{80.0} \times 15.0\right)}{15.0 + 20.0 + 0.5} \times 3.0$	C1
		$= \frac{\left(\frac{63.1}{80.0} \times 15.0\right)}{15.0 + 20.0 + 0.5} = 0.9998 \text{ V}$	A1
	(ii)	$V_{AJ} = \frac{\left(\frac{12.5}{80.0} \times 15.0\right)}{15.0 + 0.5} \times 3.0$ $= \frac{\left(\frac{12.5}{80.0} \times 15.0\right)}{15.0 + 0.5} \times 3.0$	C1

$$V_{AJ} = \frac{5.0}{5.0 + r} \times 1.00$$

C1

By solving, $r = 6.0 \, \Omega$

A1